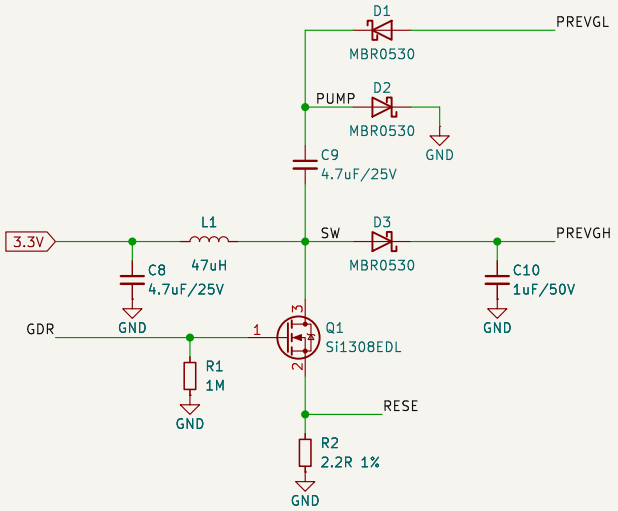
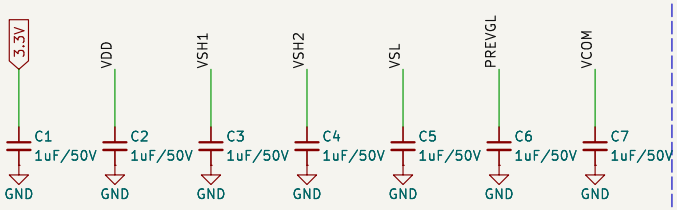
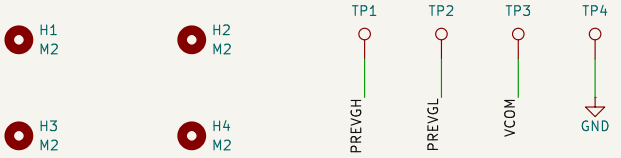
Boost: VGH/VGL generation (GDEY0213B74 reference circuit)



Panel decoupling (X5R/X7R, >=25V, close to EP_FPC1)

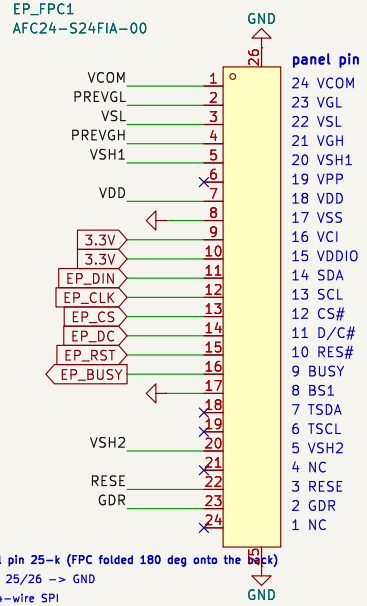


Mounting & test points



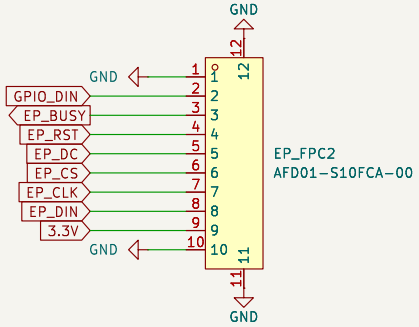
H1–H4: M2 screws + standoffs onto the 18650 holder. Back-side parts must stay below the standoff height (EP_FPC1 = 2.0mm).

EP_FPC1: panel FPC 24P 0.5mm (back side)



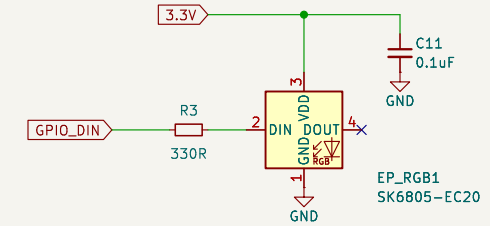
Pad k = panel pin 25−k (FPC folded 180 deg onto the back)
Mounting pads 25/26 -> GND
BS1 = L -> 4-wire SPI

EP_FPC2: to keyboard (e−paper/2.13inch−socket), 10P 0.5mm



Back side, bottom (tail) end of the upright board, centred: the same board fits the left and the right half

EP_RGB1: status LED, front, centred below the panel (SK6805−EC20)



NOTES
1. GDEY0213B74: SSD1680, 250x122, VCI 2.2–3.7V (3.3V here). BS1 = GND -> 4-wire SPI, so EP_DC is required.
2. Boost/charge pump = Good Display reference circuit: L1 47uH (Isat >= 0.4A), Q1 Si1308EDL (alt. AO3400A, SOT-23), D1–D3 MBR0530, R1 1M gate pull-down, R2 2.2 ohm RESE sense. (2.2 ohm + 47uH suit SSD ICs; UC ICs use 0.47 ohm + 10uH.)
3. Caps on PREVGH/PREVGL/VSH/VSL/VCOM must be rated >= 25V (1uF/50V X5R; C8/C9 4.7uF/25V X5R).
4. Pins 1/4 (NC), 6/7 (TSCL/TSDA), 19 (VPP) are left open.
5. Stack-up: panel taped to the FRONT, every part on the BACK. The panel FPC wraps 180 deg around the board edge into EP_FPC1, so its gold fingers face AWAY from the PCB: EP_FPC1 must be top- or dual-contact. AFC24-S24FIA-00 is dual-contact, front-insert / rear-flip: the FPC enters on the side opposite the pads, so face that side to the edge. A bottom-contact socket (e.g. AFC01) won't mate. Because of the fold, connector pad k = panel pin 25-k.
6. EP_FPC2 mates e-paper/2.13inch-socket pin for pin. Upright board: panel FPC fold at the top, EP_FPC2 on the back at the bottom (tail) end, centred, FFC exiting past the 18650: fits both keyboard halves.
7. Keep the L1/Q1/R2/C8 loop tight; route PREVGH/PREVGL/VCOM short and away from EP_CLK/EP_DIN; solid GND on the back.
8. EP_RGB1 is on the FRONT, centred below the panel; R3 (data series R, may be 0R) and C11 stay on the back. SK6805: 3.5–5.5V.

Keyboard link: EP_FPC2 10P 0.5mm FFC (reverse contacts) to e−paper/2.13inch−socket, pin k = pin
Panel + EP_RGB1 on front (panel on double−sided tape), other parts on back, M2 standoffs above 18

ErgoCai

Sheet: /
File: 2.13inch.kicad_sch

Title: 2.13inch e−Paper adapter − GDEY0213B74

Size: A4	Date: 2026-11-11	**Rev: 1.0**
KiCad E.D.A. 9.0.2		Id: 1/1